WEEK 4 · LECTURE L1 · UNIT II

HRIS: The System of Record

What an HR information system is, why one official employee record must come first, how to choose and run one, and the hidden cost of integration.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Define an HRIS as a sociotechnical system, and distinguish HRIS, HRMS and HCM suites, and HRIS from eHRM.
2. Explain the system of record, master data management, and the internal layers of one HRIS.
3. Evaluate suite against best of breed, build against buy or configure, and total cost of ownership.
4. Explain what four decades of eHRM research show, and apply DeLone and McLean, Williamson, Trist and Bamforth, and Lepak and Snell.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcome: CLO 2.

COMMIT BEFORE THEORY

Commit to an answer first

A Pune manufacturer keeps employee data in two HR systems and several spreadsheets. An employee updates her bank account in one of them. What is the deepest problem?

A

HR staff need more training in careful data entry

B

No single system is the agreed source of truth, so every connected system inherits the disagreement

C

The older system should be switched off tomorrow morning

D

The firm needs an analytics tool to detect the mismatches

EXAM LENS

The answer is B. When the CFO asks how many people work here and three systems give three answers, there is no system of record.

People, procedures, data and technology

Definition (Kavanagh and Johnson)

A system used to acquire, store, manipulate, analyze, retrieve and distribute information about an organization's people, to support HRM and managerial decisions.

Three parts

People: HR staff, managers and employees who enter, check and use the data. Procedures and policies: forms, approvals and rules. Data and technology: the database, software and devices.

HRIS and eHRM

eHRM is a way of delivering HR services through technology, to employees and managers directly; the HRIS is the technology underneath. Bondarouk et al. (2017): the field moved from automating the HR department to automating services for everyone.

SOCIOTECHNICAL VIEW

Trist and Bamforth (1951): a technical system works only when it is designed together with the social system that runs it.

RECENT EVIDENCE · STROHMEIER (2020)

Strohmeier separates the digitization, the digitalization, the digital transformation and the digital disruption of HRM: four distinct concepts, not four words for one thing.

WHERE HR'S TIME GOES

Why an HRIS matters

Kind of HR work	Examples	Share of HR time
Transactional	Record keeping, payroll entry, benefits administration	65 to 75%
Traditional	Planning, recruiting, selection, training, pay, performance	15 to 30%
Transformational	Culture change, restructuring, strategic redirection	5 to 15%

EXAM LENS

The case for an HRIS is time moved out of the first band, not software bought. A caution: Gardner, Lepak and Bartol (2003) found electronic HR often replaced administrative duties with technology related ones instead of freeing time.

Shares from Kavanagh and Johnson (2018), Chapter 1, citing Wright and colleagues; Gardner et al. as reviewed by Bondarouk et al. (2017).

HISTORY

Six eras, and three generations, of HR information systems

Era	What defined it
Prewar	Record keeping under scientific management
1945 to 1960	Job classification and unions
1963 to 1980	Employment law and compliance reporting
1980 to early 1990s	Client server systems and cost justification
1990 to 2010	ERP suites, strategic HRM, best of breed
2010 onward	Cloud and mobile

EXAM LENS

The teaching note’s three generations: standalone personnel databases (1960s and 1970s); client server systems adding talent functions (1980s and 1990s); cloud, mobile, AI and APIs built from modular services (now). Each era left systems behind: legacy, spreadsheet and cloud now live side by side.

Kavanagh and Johnson (2018), Chapter 1; course’s Unit teaching note.

SCOPE

HRIS, HRMS and HCM are steps in scope

Term	What it covers	Question that places a system
HRIS	The core record: who works here, in what role, reporting to whom	Does it hold the one authoritative employee record?
HRMS	The record plus transactions: payroll, leave, attendance	Does it run payroll, leave and attendance on that record?
HCM suite	Transactions plus talent: hiring, learning, performance, analytics	Does it run talent and analytics on the same data?

EXAM LENS

Use the terms precisely. Vendors use them loosely; examiners do not. Types of system also include transaction processing, management information, decision support, expert systems and ERP.

TWO LAYER IDEAS YOU MUST NOT CONFUSE

Inside one system, and roles across the stack

Inside one HRIS

Presentation layer: portals, mobile apps, chat; usability drives adoption and data quality. Business logic layer: rules, approvals, workflows, configurable without programming. Data layer: master data, transactions, history; the data model decides what can ever be analyzed.

Roles across the stack

Record: the one official source of employee data. Engagement: what employees touch, such as self service and chatbots. Insight: turning data into analysis and decisions.

CITE CORRECTLY

Moore (2011) contrasted systems of record with systems of engagement; insight comes from later industry work. Kavanagh and Johnson do not use these terms: cite the three roles as the course’s framing.

THE SYSTEM OF RECORD

One record, many subscribers: master data management

1

One authoritative source for each data item

2

A named owner and data steward for each item

3

Shared definitions: who counts as an employee, what a joining date is

4

Other systems read the record; they do not overwrite it

5

Every change is logged and traceable

The tell tale sign

Universal spreadsheet use alongside low satisfaction: people fall back on what they control when no system is trusted.

THE ARCHITECTURE CHOICE

Suite or best of breed

	Integrated suite	Best of breed
Strengths	One vendor and data model; integration built in; consistent reporting; one contract	Strongest tool for each need; each module replaceable on its own
Weaknesses	Some modules weaker; upgrades on the vendor’s timetable; costly to leave	Every pair of tools needs a connector; definitions drift; someone owns integration forever
Works when	Processes are standard and integration capacity is low	One core system holds the record and every specialist tool reads from it

EXAM LENS

There is no universally right answer: state the conditions. Many firms reach best of breed by accident, through acquisitions.

THE HIDDEN COST

Integration: the cost missing from every invoice

Point to point

With n systems, up to n(n − 1) ÷ 2 links: 45 for 10 systems, 120 for 16, 190 for 20. The count rises with the square.

Hub and spoke

Every system connects only to a hub, normally the system of record or an integration platform: about n − 1 links.

Three ways to integrate

Batch transfer: simple, but out of date between runs. Real time APIs: a change reaches payroll at once, but failures can cascade. Process integration: an accepted offer creates the record, accounts and learning path.

EXAM LENS

License spend is visible to the CFO; reconciliation and double entry are hidden in HR staff time. Every integration point is also a possible leak: authentication, encryption and audit logs, with DPDP duties when data flow to vendors or abroad.

SOURCING THE SYSTEM

Build, buy or configure

Route	Fits when	Main risk
Build	The process is a source of advantage, or the users are unusual (vernacular, phone only)	Dependence on a small team; slow delivery; security work you must do yourself
Buy	The process is standard, such as payroll	Fit gaps, lock in, data held abroad
Configure	A bought platform needs local fit: what most suite rollouts really do	Customization creep: every vendor upgrade breaks custom code

EXAM LENS

Bondarouk et al. (2017): in house development showed no effect on user satisfaction. Owning the code does not make people use it.

TOTAL COST OF OWNERSHIP

The license is the tip

Beneath the license fee

Implementation and configuration · data migration and cleaning · integration build and upkeep · training and change management · staff time lost to reconciliation

A worked example (Course framing)

Pilot quote: ₹30,00,000 a year for 500 seats.
₹30,00,000 ÷ 500 = ₹6,000 a seat.
₹6,000 × 6,000 seats = ₹3.6 crore a year.
Show the negotiated rate; do not assume it.

EXAM LENS

Compare options at full scale and full cost, never on the license line alone.

SELECTING, IMPLEMENTING AND GOVERNING

From selection to replacement

1 · Select

Requirements across function, technology (scale, security, integration), usability and vendor; weighted criteria, demonstrations, reference checks.

2 · Implement

Six workstreams: configuration, data migration, integration, testing, training, change management. The technical work is only part.

3 · Govern

Decision rights over configuration, request prioritization, data standards and access, often through an HR technology steering committee.

4 · Reassess

Platforms age and vendors consolidate: plan periodic reassessment and replacement.

Course’s Unit teaching note.

BONDAROUK, PARRY AND FURTMUELLER (2017)

What four decades of eHRM research show

Technology

Usefulness, ease of use, data integrity, system integration, in house or outsourced development.

Organization

Size, planning, management support, data access and privacy, budget, strategy and culture.

People

Top management support, user acceptance, HR skills, leadership: the decisive group.

EVIDENCE

A review of 69 studies from 1970 to 2010, identifying 168 adoption factors and 95 consequence factors. Efficiency is the best evidenced outcome; strategic gains are thinner. Marler and Fisher (2013) found very little systematic evidence that eHRM leads to strategic outcomes, and burden often shifts to employees and line managers. The same software produces different results in different firms.

RECENT EVIDENCE · MALIK, BUDHWAR AND KAZMI (2023)

A systematic review of 67 studies of AI in HRM sets out its organizational and employee outcomes and a strategic framework for adopting it. The research frontier has moved from eHRM to AI assisted HRM.

THEORY IN DEPTH · DELONE AND MCLEAN (1992; 2003)

What makes an information system succeed

The core idea

Success is a chain: system, information and service quality lead to use and satisfaction, which lead to net benefits.

How it works

Quality dimensions drive both use and satisfaction; together they produce benefits for individuals and the organization, which feed back into use.

Limits and critique

Use is ambiguous when it is mandatory: forced use says little about success.

IN PRACTICE

Payroll systems show near universal use because everyone must be paid; satisfaction is the better signal there.

EXAM LENS

Distinguish voluntary from mandatory use when evaluating any HR system’s success.

RECENT EVIDENCE · PETTER, DELONE AND MCLEAN (2013)

From more than 140 studies, the authors identify 43 variables that drive the success dimensions in their model: the model’s own evidence update.

THEORY IN DEPTH · WILLIAMSON (1985)

Build, buy or configure is a transaction cost choice

The core idea

Firms decide whether to make or buy by comparing the costs of organizing each transaction.

How it works

High asset specificity and uncertainty favor making in house; standard needs favor buying.

Limits and critique

It underplays capabilities and learning, which can matter more than transaction costs.

IN PRACTICE

Payroll is standard, so buy it; a vernacular frontline interface fitted to a specific workforce is specific, so configure or build.

EXAM LENS

Use asset specificity as the deciding concept; then add the capability critique.

THEORY IN DEPTH · TRIST AND BAMFORTH (1951)

Optimize the social and technical systems together

The core idea

In British coal mines, new machinery broke up work groups and productivity fell.

How it works

Joint optimization: design technology and work organization together, not one after the other.

Limits and critique

Some argue it suits industrial work more than knowledge work, though its core point travels well.

IN PRACTICE

An HRIS rollout that ignores how frontline managers actually work will be worked around.

EXAM LENS

Sociotechnical theory is the classic argument against treating HR technology as purely technical.

THEORY IN DEPTH · LEPAK AND SNELL (1998)

Information technology can change HR in three ways

The core idea

Technology can have operational, relational and transformational effects on HR.

How it works

Operational: automate transactions.
Relational: connect people.
Transformational: change HR's role.

Limits and critique

Most organizations stop at operational gains, which are easiest to measure.

IN PRACTICE

A business case that reaches only operational effects produces a cheaper paper process, not a transformation.

EXAM LENS

Classify the claimed benefits of any HR system into the three levels.

THE INDIAN SETTING

What an Indian HRIS must support

DPDP principles

Purpose limitation, data minimization, storage limitation, accuracy; access, correction, erasure and grievance redress the HRIS must support.

Labor law reporting

Provident fund (EPFO), state insurance (ESIC) and shops and establishments rules require statutory reports and retention.

Culture and diversity

Multiple languages, hierarchy in approval workflows, and gig and other nonstandard arrangements the HRIS must represent.

Course’s Unit teaching note.

APPLY

Four acquisitions, five employee lists

Teaching case (Course framing)

A Kanpur textile group has bought four smaller mills in ten years. Each mill kept its own spreadsheet of employees; head office runs an on premises HRIS bought in 2012. A cloud suite is being piloted for 300 managers. Payroll and the HRIS disagree on headcount every month, and the provident fund return is reconciled by hand.

Work it through

1. Which system should become the record, and what evidence decides it?
2. Suite or best of breed for this group, and why?
3. Who does the double entry during parallel running, and how will long serving HR officers be brought along?
4. Which TOP factor is most likely to sink the project?

A FRAMEWORK FOR ANY CASE

For any real Indian HRIS case ask five questions: what problem, and who defined it? Bought, built or configured? What happened to the record and integration? Who did not adopt, and why? What changed in outcomes, beyond activity?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
An HRIS is a piece of software	People, procedures, data and technology working together
The system with more users should become the record	Choose on data quality, integration, reach and total cost; heavy use can mean there is no alternative
A cheaper license means a cheaper system	Compare total cost at full scale
Record, engagement and insight come from Kavanagh and Johnson	They are the course’s framing, building on Moore (2011)

COMPARE AND EVALUATE

Six lenses on an HRIS decision

Theory	Use it to	Main weakness
Trist and Bamforth (1951)	Design social and technical together	Industrial origins
Lepak and Snell (1998)	Classify benefits: operational, relational, transformational	Most cases stop at operational
Williamson (1985)	Decide build or buy by asset specificity	Underplays capabilities
DeLone and McLean (1992; 2003)	Judge success beyond use	Mandatory use muddies it
Bondarouk et al. (2017)	Explain adoption by TOP factors	Thin evidence on strategic outcomes
Master data management	Keep one record	Needs governance, not only technology *

* Course framing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (1 of 2)

A Kolkata FMCG company runs two HR systems that both hold employee records, plus spreadsheets kept by each plant. Using the system of record idea, the suite versus best of breed choice and the TOP framework, advise the CHRO on how to consolidate, and state one limit of your advice. **20 marks**

A strong answer: defines an HRIS as sociotechnical and the record as one source governed by master data management; compares the two systems on evidence; chooses suite or best of breed with conditions; applies TOP and notes a limit.

Critically evaluate the choice between an integrated HCM suite and a best of breed architecture for a large Indian organization with a dispersed frontline workforce. **20 marks**

A strong answer: defines both; compares on integration, fit, lock in and data residency; uses transaction cost logic; concludes conditionally around the system of record.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (2 of 2)

“The same HR software produces different results in different firms.” Discuss with reference to eHRM research and sociotechnical theory. **20 marks**

A strong answer: summarizes Bondarouk et al.’s three factor groups; uses Trist and Bamforth; distinguishes operational from transformational outcomes (Lepak and Snell).

Explain the DeLone and McLean model and why mandatory use complicates it. **10 marks**

A strong answer: states the chain from quality to benefits and explains why use is a weak measure when mandatory.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **One employee, one record.**
   A system of record is the foundation every other tool depends on.

2. **Integration is the hidden cost.**
   Compare options at full scale and full cost.

3. **An HRIS is sociotechnical.**
   Design the work and the system together.

4. **Efficiency is proven; transformation is not automatic.**
   Implementation and acceptance decide the outcome.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Bondarouk, T., Parry, E., & Furtmueller, E. (2017). Electronic HRM: Four decades of research on adoption and consequences. *IJHRM, 28*(1), 98 to 131.
- Johnson, R. D., Carlson, K. D., & Kavanagh, M. J. (2025), Chapter 1 (the evolution of HRM and HRIS; the HRIS architecture) and Chapter 2 (electronic HRM).
- An Indian HRIS or ATS implementation case.

> **NOTE**
> The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Bondarouk, T., Parry, E., & Furtmueller, E. (2017). Electronic HRM: Four decades of research on adoption and consequences. *The International Journal of Human Resource Management, 28*(1), 98 to 131. https://doi.org/10.1080/09585192.2016.1245672
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